

14-795 Challenge Analysis

Defending Against Indirect Prompt Injections: A Multi-Layered Verification Pipeline

Privacy

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Strengths: Privacy

- Local Architecture:
 - We use Ollama, a local model that runs weights on your own machine.
 - No Cloud API calls means no data retention risks
 - Data processed locally, only outputs are shared
- Structure Data Sanitization:
 - Layer 1 delimits untrusted inputs with (EXTERNAL_DATA_START/END), this limits the attack surface, as the model sees as the “instruction”
 - Layer 2 terminates processing early when malicious content is detected and stops processing immediately. (Attacker payload never reaches reasoning model)
 - Fail fast and minimize the data flow

Weaknesses: Privacy

- **Lack of Formal E-Guarantee:**
 - No mathematical **Privacy Budget** to limit information leakage over time.
 - Repeated queries can allow an attacker to **probe** and reconstruct private data.
- **Linkage Attack Vulnerability:**
 - Sanitized logs can be de-anonymized (Ex.. by timestamps)
 - Similar to the Netflix Prize breach where external data unmasked "private" users.
- **Raw Data Sensitivity:**
 - Internal layers still operate on un-noised financial/PII data.
 - No "Local Differential Privacy" to protect data before it enters the pipeline.

Opportunities for improvement

- **Differential Privacy Integration:**
 - Apply Laplacian Noise to outputs to provide a formal privacy guarantee.
 - Implement an E budget to track and limit total privacy loss per user.
 - **Benefit:** Provides a formal mathematical guarantee that prevents attackers from reconstructing raw data through repeated queries.
- **Input Noise Injection:**
 - Add Local Differential Privacy
 - Add noise to raw data locally *before* processing to ensure zero raw data exposure.
 - **Benefit:** even if the pipeline or logs are compromised, the source data remains mathematically obscured
- **Automated Data Masking:**
 - Programmatic redaction of PII (account numbers, addresses) in all Layers.
 - **Benefit:** Prevents sensitive "outliers" from being recorded in reasoning logs, reducing the risk of re-identification via linkage attacks.